Biology Class XI - Full Length Test

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Instructions

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Sections: **1** Time: **45 Minutes**

Section: **BIology**

1. The taxonomic unit 'Phylum' in the classification of animals is equivalent to which hierarchical level in classification of plants

(a) Class
(b) order
(c) Division
(d) Family

Answer key : c

Correct Marks : 4 Wrong Marks : -1

2. By whom was the five kingdom classification of living beings presented?

(a) Aristotle
(b) R.H. Whittaker
(c) Carl woese
(d) John Ray

Answer key : b

Correct Marks : 4 Wrong Marks : -1

3. In Magnifera indica, the word Mangifera is a

(a) Genus
(b) Species
(c) Variety
(d) Order

Answer key : a

Correct Marks : 4 Wrong Marks : -1

4. In majority of higher animals and plants, reproduction and growth are

(a) Mutual exclusive events
(b) Synonymous events
(c) Synonymous events during in vitro culture
(d) Repulsive events

Answer key : a

Correct Marks : 4 Wrong Marks : -1

5. Group of organisms in which cell wall forms two thin overlapping shells are

(a) Responsible for bioluminescence.

(b) Chief producers of ocean

(c) Prokaryotes

(d) Heterotrophs.

Answer key : b

Correct Marks : 4 Wrong Marks : -1

6. The fruiting bodies of Agaricus are also known as

(a) Cleistothecium

(b) Fairy rings

(c) Basidiocarp

(d) Ascocarp

Answer key : c

Correct Marks : 4 Wrong Marks : -1

7. Which of the following does not belong to the class basidiomycetes?

(a) Agaricus

(b) Ustilago

(c) Alternaria

(d) Puccinia

Answer key : c

Correct Marks : 4 Wrong Marks : -1

8. 'Peptidoglycan' is a characteristic constituent of the cell wall of

(a) Eubacteria and unicellular eukaryotes

(b) Bacteria and cyanobacteria

(c) Archaebacteria and eukaryotes

(d) All members of 'monera' and 'protista'

Answer key : b

Correct Marks : 4 Wrong Marks : -1

9. The smallest organism that is capable of autonomous growth and reproduction is

(a) Virus

(b) Viroid

(c) Mycoplasma

(d) None of these

Answer key : c

Correct Marks : 4 Wrong Marks : -1

10. The causative agent of mad-cow disease is a

(a) Bacterium
(b) Prion
(c) Worm
(d) Virus

Answer key : b

Correct Marks : 4 Wrong Marks : -1

11. The vascular tissue is absent in

(a) Algae, fungi and pteridophytes
(b) Thallophytes and bryophytes
(c) Bryophytes and pteridophytes
(d) Angiosperm and gymnosperm

Answer key : b

Correct Marks : 4 Wrong Marks : -1

12. Which of the following are examples of unicellular algae?

(a) (1) Chlamydomonas
(b) (2) Ricca
(c) (3) Volvox
(d) Both (1) & (3)

Answer key : a

Correct Marks : 4 Wrong Marks : -1

13. Phaeophyceae are _____ algae

(a) Green
(b) Red
(c) Brown
(d) Blue

Answer key : c

Correct Marks : 4 Wrong Marks : -1

14. Double fertilization is present in

(a) Pinus
(b) Pteris
(c) Eucalyptus
(d) Fucus

Answer key : c

Correct Marks : 4 Wrong Marks : -1

15. What is common in all the three, Funaria, Dryopteris and Ginkgo?

(a) Presence of archegonia
(b) Well-developed vascular tissues
(c) Independent gametophyte
(d) Independent sporophyte

Answer key : a

Correct Marks : 4 Wrong Marks : -1

16. In diploblastic animals, cells are arranged into two embryonic layers.

(a) TRUE

(b) FALSE

Answer key : a

Correct Marks : 4 Wrong Marks : -1

17. Triploblastic and coelomate animals with complete gut and closed circulatory system belongs to a phylum whose member is:

(a) Ascaris

(b) Bombyx

(c) Nereis

(d) Taenia

Answer key : c

Correct Marks : 4 Wrong Marks : -1

18. An egg laying mammal is

(a) Didelphys

(b) Macaca

(c) Ornithorhynchus

(d) Macropus

Answer key : c

Correct Marks : 4 Wrong Marks : -1

19. Birds are not characterized by:

(a) Forelimbs modified into wings

(b) Fully ossified endoskeleton

(c) Skin dry without glands except for oil gland

(d) Oviparous and external fertilization

Answer key : d

Correct Marks : 4 Wrong Marks : -1

20. Which of the following is not an Echinoderm character?

(a) Exclusively marine

(b) Radial symmetry in adults

(c) Protostome embryonic development

(d) Water vascular system (i) (v) (iv)

Answer key : c

Correct Marks : 4 Wrong Marks : -1

21. The wheat grain has an embryo with one large, shield-shaped cotyledon known as?

(a) epiblast
(b) coleorrhiza
(c) scutellum
(d) coleoptile

Answer key : c

Correct Marks : 4 Wrong Marks : -1

22. Rachis is

(a) A prominent vein
(b) Midrib
(c) Found in pinnately compound leaf
(d) All of the above

Answer key : d

Correct Marks : 4 Wrong Marks : -1

23. Drupe fruit develops from

(a) Monocarpellary superior ovaries and are one seeded
(b) Monocarpellary inferior ovaries and are many seeded
(c) Monocarpellary superior ovaries and are many seeded
(d) Bicarpellary superior ovaries and are many seeded

Answer key : a

Correct Marks : 4 Wrong Marks : -1

24. In coconut fruit, the hard shell is

(a) Endocarp
(b) Fused structure of mesocarp and endocarp
(c) Fused structure of epicarp and mesocarp
(d) Epicarp

Answer key : a

Correct Marks : 4 Wrong Marks : -1

25. Non-albuminous seed is produced in

(a) Maize
(b) Castor
(c) Wheat
(d) Pea

Answer key : d

Correct Marks : 4 Wrong Marks : -1

26. The stomata in an isobilateral leaf

(a) Are present only on the adaxial epidermis

(b) Are present only on the abaxial epidermis

(c) Are absent on both the surfaces of the epidermis

(d) Are present on both the surfaces of the epidermis

Answer key : d

Correct Marks : 4 Wrong Marks : -1

27. Which of the following are example of lateral meristem?

(a) Fascicular vascular cambium

(b) Interfascicular cambium

(c) Cork cambium

(d) All

Answer key : d

Correct Marks : 4 Wrong Marks : -1

28. Casparian strips are barrel shaped cells in which tangential and radial walls are impermeableto water because of

(a) suberin deposition

(b) cellulose deposition

(c) cutin deposition

(d) casparin deposition

Answer key : a

Correct Marks : 4 Wrong Marks : -1

29. Root hair develop from the region of

(a) elongation

(b) root cap

(c) meristematic activity

(d) maturation.

Answer key : d

Correct Marks : 4 Wrong Marks : -1

30. Phloem in gymnosperms lack

(a) both sieve tubes and companion cells

(b) albuminous cells and sieve cells

(c) sieve tubes only

(d) companion cells only

Answer key : a

Correct Marks : 4 Wrong Marks : -1

31. The oocytes of some vertebrates get arrested for years or months in:

(a) Telophase - I
(b) Diplotene
(c) Diakinesis
(d) Metaphase - I

Answer key : b

Correct Marks : 4 Wrong Marks : -1

32. Select the correct option based on statements below:
Assertion (A): Meiosis leads to recombinations of genes.
Reason (R): Crossing over occurs in pachytene sub-stage of prophase I of meiosis.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(c) (A) is true but (R) is false.
(d) Both (A) and (R) are false.

Answer key : a

Correct Marks : 4 Wrong Marks : -1

33. In prophase I of Meiosis, chromosomes start pairing together and synapsis takes place. This process occurs during which of the following stage?

(a) Pachytene
(b) Diplotene
(c) Leptotene
(d) Zygotene

Answer key : d

Correct Marks : 4 Wrong Marks : -1

34. The centriole undergoes duplication during:

(a) Prophase
(b) Metaphase
(c) G_2 phase
(d) S-phase

Answer key : d

Correct Marks : 4 Wrong Marks : -1

35. Which one of the following never occurs during mitotic cell division?

(a) Coiling and condensation of the chromatids
(b) Spindle fibres attach to kinetochores of chromosomes
(c) Movement of centrioles towards opposite poles
(d) Pairing of homologous chromosomes

Answer key : d

Correct Marks : 4 Wrong Marks : -1

36. Single membrane bound organelles are

(a) Lysosomes
(b) Chloroplasts
(c) Peroxisomes
(d) Both (1) and (3)

Answer key : d

Correct Marks : 4 Wrong Marks : -1

37. What is the arrangement of microtubules in cilia and flagella?

(a) $9+0$
(b) $9+2$
(c) 9
(d) Both (1) and (3)

Answer key : b

Correct Marks : 4 Wrong Marks : -1

38. Extra nuclear DNA is found in which two cell organelles?

(a) Mitochondria & ER
(b) Ribosomes and ER
(c) Chloroplasts & mitochondria
(d) Chloroplast and ribosomes

Answer key : c

Correct Marks : 4 Wrong Marks : -1

39. The green colour plastid

(a) Leucoplast
(b) Chromoplast
(c) Chloroplast
(d) None of the above

Answer key : c

Correct Marks : 4 Wrong Marks : -1

40. A semi permiable membrane -

(a) Cell wall
(b) Cell membrane
(c) tonoplast
(d) All of the above

Answer key : b

Correct Marks : 4 Wrong Marks : -1

41. Prosthetic groups differ from co - enzymes in that

(a) They require metal ions for their activity
(b) They (prosthetic groups) are tightly bound to apoenzymes.
(c) Their association with apoenzymes is transient.
(d) They can serve as co - factors in a number of enzyme - catalysed reactions.

Answer key : b

Correct Marks : 4 Wrong Marks : -1

42. Fats are

(a) Ester of fatty acid and aldehyde

(b) Ester of fatty acid only

(c) Glycerol only

(d) Esters of fatty acid with glycerol

Answer key : d

Correct Marks : 4 Wrong Marks : -1

43. A nucleoside differs from nucleotide is not having

(a) Sugar

(b) Nitrogen

(c) Phosphate

(d) phosphate and sugar

Answer key : c

Correct Marks : 4 Wrong Marks : -1

44. Amino acid is a:

(a) Substituted methane

(b) Substituted ethane

(c) Any acid having amino group

(d) Derivative of indol acetic acid

Answer key : a

Correct Marks : 4 Wrong Marks : -1

45. Valine is a :

(a) Simple fatty acid

(b) Complex fatty acid

(c) Acidic amino acid

(d) Neutral amino acid

Answer key : d

Correct Marks : 4 Wrong Marks : -1

46. The binding of Hb with oxygen forms

(a) Methamoglobin

(b) Carbhaminohaemoglobin

(c) Oxyhaemoglobin

(d) Carbaminohaemoglobin

Answer key : c

Correct Marks : 4 Wrong Marks : -1

47. During inspiration which of the following events takes place

(a) Diaphragm and external intercostal muscles contracts

(b) Diaphragm and internal intercostal muscles relax

(c) Intra Pleural pressure is less than atmospheric pressure

(d) Both (A) and (C)

Answer key : d

Correct Marks : 4 Wrong Marks : -1

48. The partial pressure of CO2 is minimum in the

(a) Atmospheric air

(b) Alveoli

(c) Deoxygenated blood

(d) Oxygenated blood

Answer key : a

Correct Marks : 4 Wrong Marks : -1

49. Which of the following is not supported by cartilaginous rings

(a) Primary bronchioles

(b) Tertiary bronchioles

(c) Tracheae

(d) Alveoli

Answer key : d

Correct Marks : 4 Wrong Marks : -1

50. Diffusion membrane consists of

(a) Squamous epithelium of alveoli

(b) Endothelium of alveolar capillaries

(c) Basement substance between them

(d) All of the above

Answer key : d

Correct Marks : 4 Wrong Marks : -1

51. Angina pectoris is a major symptom of

(a) Myocardial infarction

(b) Cyanosis

(c) High blood pressure

(d) Low blood pressure

Answer key : a

Correct Marks : 4 Wrong Marks : -1

52. A person with AB blood group:

(a) Has A and B antigens in his plasma

(b) Lacks any antibodies on RBC surface

(c) Can receive blood transfusion from a person of his own group.

(d) Can donate blood to all other group recipients

Answer key : c

Correct Marks : 4 Wrong Marks : -1

53. To avoid Erythroblastosis Foetalis -

(a) The mother is administered anti-Rh antibodies.

(b) The baby is administered Rh antibodies.

(c) The mother is administered Rh antibodies

(d) The baby is administered anti - Rh antibodies

Answer key : a

Correct Marks : 4 Wrong Marks : -1

54. The lumen of artery is _____ than vein.

(a) Narrower

(b) Wider

(c) Same

(d) Very much reduced

Answer key : a

Correct Marks : 4 Wrong Marks : -1

55. What percentage of ventricular filling is achieved by atrial contraction?

(a) 30

(b) 50

(c) 70

(d) 90

Answer key : a

Correct Marks : 4 Wrong Marks : -1

56. Which of the following statements are correct?

A: An excessive loss of body fluid from the body switches off osmoreceptors.

B: ADH facilitates water reabsorption to prevent diuresis.

C: ANF causes vasodilation.

D: ADH causes increase in blood pressure.

E: ADH is responsible for decrease in GFR.

Choose the correct answer from the options given below:

(a) **C**, **D** and **E** only

(b) **A** and **B** only

(c) **B**, **C** and **D** only

(d) **A**, **B** and **E** only

Answer key : c

Correct Marks : 4 Wrong Marks : -1

57. Given below are statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Nephrons are of two types: Cortical & Juxta medullary, based on their relative position in cortex and medulla.
Reason (R): Juxta medullary nephrons have short-loop of Henle whereas, cortical nephrons have longer loop of Henle.

In the light of the above statements, choose the correct answer from the options given below:

(a) **(A)** is False, **(R)** is True.

(b) Both **(A)** and **(R)** are True and **(R)** correctly explains **(A)**.

(c) Both **(A)** and **(R)** are True but **(R)** does not correctly explain**(A)**.

(d) **(A)** is True, **(R)** is False.

Answer key : d

Correct Marks : 4 Wrong Marks : -1

58. Choose the odd one out:

(a) Afferent arteriole, efferent arteriole, loop of Henle vasa recta

(b) Proximal convoluted tubule, distal convoluted tubule, loop of Henle

(c) Renal pelvis, renal papilla, medullary pyramid

(d) Proximal convoluted tubule, loop of Henle, distal convoluted tubule, collecting tubule

Answer key : a

Correct Marks : 4 Wrong Marks : -1

59. Which of the following is not a part of medullary pyramid?

(a) Proximal convoluted tubule

(b) Distal convoluted tubule

(c) Malpighian corpuscle

(d) More than one of the above

Answer key : d

Correct Marks : 4 Wrong Marks : -1

60. **Assertion:** The urinary bladder stretches as urine collects
Reason: The urinary bladder is lined by transitional epithelium

(a) If both assertion and reason are true and reason is the correct explanation of the assertion.

(b) If both assertion and reason are true but the reason is not the correct explanation of assertion

(c) If assertion is true but reason is false

(d) If both assertion and reason are false

Answer key : a

Correct Marks : 4 Wrong Marks : -1

61. The reflex action is under the control of

(a) Motor nerves
(b) Sensory nerves
(c) Central nervous system
(d) Sympathetic nervous system

Answer key : c

Correct Marks : 4 Wrong Marks : -1

62. The nerves leading to the central nervous system are called

(a) Efferent nerves
(b) Afferent nerves
(c) Both (1) and (2)
(d) None of the above

Answer key : b

Correct Marks : 4 Wrong Marks : -1

63. 'Pons' connects the

(a) Two cerebral hemispheres
(b) Two lobes of cerebellum
(c) Cerebrum and cerebellum
(d) Spinal cord with medulla oblongata

Answer key : c

Correct Marks : 4 Wrong Marks : -1

64. A thirsty person drinking water and would continue drinking till he receives signals from his brain. These signals would come from which part of his brain.

(a) Cerebral hemisphere
(b) Hypothalamus
(c) Cerebellum
(d) Medulla oblongata

Answer key : b

Correct Marks : 4 Wrong Marks : -1

65. Which of the following statement is not correct about cortex of cerebrum?

(a) It consists of grey matter.
(b) It consists of white matter.
(c) It shows prominent folds.
(d) It contains motor areas, sensory areas and association areas.

Answer key : b

Correct Marks : 4 Wrong Marks : -1

66. Thymus gland releases _____ hormone

(a) T_4
(b) T_3
(c) Thymosins
(d) TCT

Answer key : c

Correct Marks : 4 Wrong Marks : -1

67. A child with weak immune system could have problem in which of the following gland

(a) Thyroid gland
(b) Parathyroid gland
(c) Thymus
(d) Pituitary gland

Answer key : c

Correct Marks : 4 Wrong Marks : -1

68. Adrenal gland is present at the

(a) lateral side of each kidney
(b) dorsal side of each kidney
(c) posterior side of each kidney
(d) anterior side of each kidney

Answer key : d

Correct Marks : 4 Wrong Marks : -1

69. Correct characteristics of insulin includes

(a) hypoglycemic hormone
(b) decreases the blood sugar
(c) act on adipose tissue and hepatocytes
(d) All of the above

Answer key : d

Correct Marks : 4 Wrong Marks : -1

70. Erythropoietin

(a) stimulates erythropoiesis
(b) inhibits erythropoiesis
(c) inhibits platelets formation
(d) stimulates platelets formation

Answer key : a

Correct Marks : 4 Wrong Marks : -1

71. Sarcolemma is the outer covering of

(a) Muscle fibre
(b) Cartilage
(c) Nerve fibre
(d) Collagen fibre

Answer key : a

Correct Marks : 4 Wrong Marks : -1

72. The human skull is

(a) Dicondylic
(b) Monocondylic
(c) Both (1) and (2)
(d) Without condyles

Answer key : a

Correct Marks : 4 Wrong Marks : -1

73. Fibrous joints are found in

(a) between flat bones of skull
(b) pubic symphysis
(c) between vertebrae
(d) all of these.

Answer key : a

Correct Marks : 4 Wrong Marks : -1

74. The 8th, 9th and 10th rib are also called

(a) Vertebrochondral ribs
(b) Vertebrostemal ribs
(c) Floating robs
(d) None of the above

Answer key : a

Correct Marks : 4 Wrong Marks : -1

75. Appendicular skeleton includes

(a) girdles and their limbs
(b) vertebrae
(c) skull and vertebral column
(d) ribs and sternum.

Answer key : a

Correct Marks : 4 Wrong Marks : -1

76. An action spectrum of photosynthesis was first described by:

(a) Jan Ingenhousz
(b) Julius von Sachs
(c) T.W. Engelmann
(d) Cornelius van Niel

Answer key : c

Correct Marks : 4 Wrong Marks : -1

77. The bundle sheath cells have

(a) Thick walls, no intercellular spaces and large number of chloroplasts
(b) Thin walls, no intercellular spaces and several chloroplasts
(c) Thick walls, many intercellular spaces and few chloroplasts
(d) Thin walls, many intercellular spaces and no chloroplasts

Answer key : a

Correct Marks : 4 Wrong Marks : -1

78. Kranz anatomy is typical of

(a) C_4 plants

(b) C_3 plants

(c) C_2 plants

(d) CAM plants

Answer key : a

Correct Marks : 4 Wrong Marks : -1

79. Stomata of CAM plants

(a) Are always open

(b) Open during the day and close at night

(c) Open during the night and close during the day

(d) Never open

Answer key : c

Correct Marks : 4 Wrong Marks : -1

80. The first stable product of Calvin cycle is

(a) 3-phosphoglycerate

(b) 1, 3-bisphophoglycerate

(c) Glyceraldehyde-3-phosphate

(d) Ribulose-5-phosphate

Answer key : a

Correct Marks : 4 Wrong Marks : -1

81. In Kreb's cycle, GTP is formed in

(a) substrate level phosphorylation

(b) oxidative phosphorylation

(c) non-cyclic photophosphorylation

(d) cyclic photophosphorylation

Answer key : a

Correct Marks : 4 Wrong Marks : -1

82. RQ (Respiratory Quotient) is defined as

(a) Volume of CO_2 evolved = volume of O_2 consumed

(b) $\frac{Volume\ of\ O_2\ consumed}{Volume\ of\ CO_2\ evolved}$

(c) $\frac{Volume\ of\ CO_2\ evolved}{Volume\ of\ O_2\ consumed}$

(d) $\frac{Volume\ of\ O_2\ evolved}{Volume\ of\ CO_2\ consumed}$

Answer key : c

Correct Marks : 4 Wrong Marks : -1

83. The energy utilizing step of glycolysis is

(a) 1st and 3rd steps
(b) 3rd step only
(c) Last step only
(d) 1st step only

Answer key : a

Correct Marks : 4 Wrong Marks : -1

84. Where does ETS occur?

(a) Inner Mitochondrial Membrane
(b) Peri Mitochondrial Space
(c) Outer Mitochondrial Membrane
(d) Inter Membrane space

Answer key : a

Correct Marks : 4 Wrong Marks : -1

85. Connecting link between glycolysis and TCA cycle is

(a) Link reaction
(b) EMP pathway
(c) ETC
(d) Citric acid cycle

Answer key : a

Correct Marks : 4 Wrong Marks : -1

86. **Assertion:** Kinetin is found naturally in plants.
Reason: ABA breaks seed and bud dormancy.

(a) If both assertion and reason are true and reason is the correct explanation of assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of assertion.
(c) If assertion is true but reason is false.
(d) If both assertion and reason are false.

Answer key : d

Correct Marks : 4 Wrong Marks : -1

87. **Assertion:** Development is the sum of growth and differentiation.
Reason: Development in plants is under the control of extrinsic factors only.

(a) If both assertion and reason are true and reason is the correct explanation of assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of assertion.
(c) If assertion is true but reason is false.
(d) If both assertion and reason are false.

Answer key : d

Correct Marks : 4 Wrong Marks : -1

88. Cytokinins are mostly

(a) Glucosides
(b) Phenolic
(c) Amino purines
(d) Organic acids.

Answer key : c

Correct Marks : 4 Wrong Marks : -1

89. Which of the following is both a growth promoter as well as a growth inhibitor?

(a) Auxin
(b) gibberellic acid
(c) ABA
(d) Ethylene

Answer key : d

Correct Marks : 4 Wrong Marks : -1

90. Abscisic acid performs all of the functions except

(a) Inhibits seed germination
(b) Stimulates closure of stomata
(c) Induces seed dormancy
(d) Control xylem differentiation

Answer key : d

Correct Marks : 4 Wrong Marks : -1